Zihan Dong

NC | puma122707@gmail.com |(704) 951-5906 |GitHub: zdong104 | LinkedIn: zihandong1004 | Google Scholar

SUMMARY

ML Engineer building long-horizon LLM agents and RL post-training systems — tool use, multimodal memory, verifiable task execution, and efficient serving. 12 papers, 3 patents, 200+ citations (HomePage).

EDUCATION

Georgia Institute of Technology – Atlanta, GA Aug 2024 - May 2026
Master of Science in Computer Science (Specialization: ML / AI / Cognitive Systems) GPA: 3.7/4.0
North Carolina State University – Raleigh, NC Aug 2020 - May 2024
Bachelor of Science in Mechanical Engineering, Minor in Business Administration GPA: 3.8/4.0

- NSF-funded researcher | Grand Challenge Scholar (VP)

PROFESSIONAL EXPERIENCE

Lenovo – AI Ecosystem Engineer (Rotation Program) *Morrisville, NC | 2025 – 2026*

- Built an end-to-end proactive agent system for long-horizon tasks — perception → memory → reasoning → tool execution → feedback — with multimodal memory and recovery, improving task success rate 23% and cutting latency 5%
- Implemented function-calling / tool-use layer enabling agents to act on external systems and APIs
- Developed eval pipeline for long-horizon agent tasks — success rate, hallucination frequency, latency, recovery behavior

Lenovo – Workstation AI Solution Engineer *Morrisville, NC | 2024 – Present*

- Designed tool-calling and CUA-based agent workflows automating system diagnostics, scheduling, and recovery.
- Built efficient LLM serving for workstation GPUs — quantization, paged KV cache, continuous batching.
- Built the RL post-training pipeline (GRPO/GSPO/DAPO) for domain-specific models — rollout generation, reward modeling, and distributed trainer integration — in collaboration with architecture and systems teams.
- Developed virtualization and hypervisor environments (e.g., Slurm, remote compute) for distributed AI workflows and model experimentation for On-device Workstation.

North Carolina State University – Interactive Intelligence Research Intern *Remote | Apr 2024 – Oct 2024*

- Built a multimodal RAG assistant for manufacturing QA over text, images, tables, and figures; used CoT/ReAct prompting and a custom context store to cut failures in long-context, multimodal reasoning
- Developed a PyTorch CV pipeline (detection → pose → motion tracking) with multi-GPU CUDA parallelization, achieving 30%+ inference-latency reduction
- Deployed models to mobile clients and cloud GPU clusters (GCP/Azure)

KPMG AI Lab – LLM & Vision Research Intern *Nanjing, China | May 2023 – Aug 2023*

- Fine-tuned LLMs for regulatory intelligence using PEFT/LoRA, cutting GPU memory cost of adaptation; trained CV models for document detection and segmentation
- Built the legal-data pipeline (collection, cleaning, structuring); benchmarked SOTA architectures for model selection

PUBLICATIONS & PATENTS

Self-Consolidating Language Models: Continual Knowledge Incorporation from Context

- Proposed SCoL, a meta-RL post-training method where an LLM learns to select which of its own layers to update for continual context consolidation, beating prompting, TTT, and sequential fine-tuning on acquisition and retention

CADWorld: A Computer-Use Benchmark for Professional Mechanical CAD Design

- Built CADWorld, a FreeCAD computer-use benchmark scoring agents on verified CAD artifacts rather than proxy goals, and benchmarked frontier + open models under a sharded multi-VM evaluation harness

Additional publications: NeurIPs, ICLR, KDD, AAAI, ASA, JIMS (see Google Scholar)
Patent: ComputerAgent: A Hierarchical Reinforcement Learning Framework for General-Purpose Computer Control (IDF-175567)
Patent: Avoid Catastrophic Forgetting with Rank-1 Fisher from Diffusion Models (ID:63/888,843)
Patent: Self-Consolidating Language Models: Continual Knowledge Incorporation from Context. (ID:2026-488)

TECHNICAL SKILLS

LLMs, Agents & RL: function calling, ReAct, hierarchical memory, long-context reasoning, agent eval, RL post-training (GRPO, DAPO, GSPO, Self-Evolve), reward design, Multimodal LLM, VLMs, RAG
Systems: on-device inference, GPU optimization, distributed training (Slurm, Ray)
Languages: English (professional), Chinese (native), Japanese (basic)